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# Operative and nonoperative outcomes of emergency general surgery conditions: An observational study using a novel instrumental variable

**Elinore J. Kaufman, MD MSHP**[1], **Luke J. Keele, PhD**[2], **Christopher J. Wirtalla, BA**[2], **Claire B. Rosen, MD**[2], **Sanford E. Roberts, MD**[2], **Catherine L. Mavroudis, MD MBA**[2], **Patrick M. Reilly, MD**[1], **Daniel N. Holena, MD MSCE**[1], **Matthew D. McHugh, PhD**[3], **Dylan Small, PhD**[4], **Rachel R. Kelz, MD, MSCE, MBA, FACS**[2]

[1]Division of Traumatology, Surgical Critical Care, and Emergency Surgery, Center for Surgery and Health Economics, University of Pennsylvania Perelman School of Medicine, The Leonard Davis Institute of Health Economics, The University of Pennsylvania

[2]Department of Surgery, Center for Surgery and Health Economics, The University of Pennsylvania Perelman School of Medicine

[3]Department of Biobehavioral Health Sciences and Center for Health Outcomes and Policy Research, The University of Pennsylvania School of Nursing

[4]Department of Statistics and Data Science, The Wharton School, The University of Pennsylvania

## Abstract

**Objective:** To determine the effect of operative versus non-operative management of emergency general surgery conditions on short- and long-term outcomes.

**Summary Background Data:** Many emergency general surgery conditions can be managed either operatively or non-operatively, but high-quality evidence to guide management decisions is scarce.

**Methods:** We included 507,677 Medicare patients treated for an emergency general surgery condition between 7/1/2015 and 6/30/2018. Operative management was compared to nonoperative management using a preference-based instrumental variable analysis and near-far matching to minimize selection bias and unmeasured confounding. Outcomes were mortality, complications, and readmissions.

**Results:** For hepatopancreaticobiliary conditions, operative management was associated with lower risk of mortality at 30 (−2.6% [95% CI −4.0, −1.3]), 90 (−4.7% [−6.50, −2.8]), and 180 (−6.4% [−8.5, −4.2]) days. Among 56,582 intestinal obstruction patients, operative management was associated with a higher risk of inpatient mortality (2.8% [0.7, 4.9]) but no significant difference thereafter. For upper gastrointestinal conditions, operative management was associated

Corresponding Author: Rachel R. Kelz, MD MSCE MBA, Department of Surgery, 3400 Spruce Street, 4 Silverstein, Philadelphia, PA 19104, Rachel.Kelz@Pennmedicine.upenn.edu, Phone: 215-662-2030, Fax: 215-662-7476.

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with a 9.7% higher risk of in-hospital mortality [6.4, 13.1], which increased over time. There was a 6.9% higher risk of inpatient mortality [3.6, 10.2] with operative management for colorectal conditions, which increased over time. For general abdominal conditions, operative management was associated with 12.2% increased risk of inpatient mortality (8.7, 15.8). This effect was attenuated at 30 days (8.5% [3.8, 13.2]) and nonsignificant thereafter.

**Conclusions:** The effect of operative emergency general surgery management varied across conditions and over time. For colorectal and upper gastrointestinal conditions, outcomes are superior with nonoperative management, whereas surgery is favored for patients with hepatopancreaticobiliary conditions. For obstructions and general abdominal conditions, results were equivalent overall. These findings may support patients, clinicians, and families making these challenging decisions.

## Mini Abstract

Many emergency general surgery conditions can be managed operatively or nonoperatively. In this national, retrospective cohort study, we found that non-operative management was beneficial for colorectal and upper GI conditions, while operative management improved survival in hepatopancreatobiliary conditions. For obstructions and general abdominal conditions, results were equivalent.

## Introduction

Surgeons are often characterized as “sometimes wrong, but never in doubt,”[1] yet decision-making in emergency general surgery can be uncertain. Emergency operations are high risk with complications up to 50%, readmissions up to 20%, and mortality up to 15%.[2–4] More than 3 in 5 emergency surgery patients are over age 65, and older adults bear higher risk of adverse outcomes.[5–7] It is often unclear which among the more than 3 million adults who are hospitalized and treated for an emergency general surgery condition in the US each year should undergo surgery, and which would benefit from nonoperative management.

Studies that clearly compare operative and non-operative management of emergency surgery conditions are scarce, and results have been mixed. For example, while appendicitis had long been considered a surgical emergency, intensive study over the last decade suggests that nonoperative management of acute appendicitis is safe in most cases.[8,9] However, results are heterogenous, follow up is limited, recurrence rates are high, and elderly patients, for whom nonoperative management may be most attractive, may be particularly ill-equipped to handle a second bout of disease.[10]

Randomized control trials are rarely possible for emergency general surgery conditions because of the invasive nature of operative treatment and the acute nature of emergency surgery. Observational studies of emergency general surgery conditions are often limited by selection bias associated with treatment decisions. To address the major gap in our understanding of how to treat patients with emergency general surgery conditions while accounting for key sources of bias, we used a matched, instrumental variable (IV) analysis to compare operative to nonoperative management of older emergency surgery patients

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in a national cohort. We hypothesized that operative management would result in higher short-term complications and mortality, but better long-term outcomes.

In a randomized controlled trial, treatment is assigned by a flip of a coin or a random number generator, ensuring that the study groups who receive the treatment of interest are otherwise as identical as possible to the control group. Without randomization, we must study groups as they were treated in reality, but in reality, groups who receive a treatment such as operative management of a given disease, may differ greatly from those who do not. Importantly, conventional techniques may not be able to measure important difference between these groups, and therefore may not be able to adjust for them. An instrumental variable can aide our understanding by serving the role of a real-life random number generator. An IV is an outside force that determines the mode of treatment, without taking into account any differences among patients. Thus the instrumental variable can create groups that are nearly as similar as if they had been determined by random chance. The instrumental variable analysis provides insights for patients in the marginal population —those whose treatment would have been affected by different levels of the instrumental variable. As such, we did not assess the effect of surgery on patients who were always, already headed to the operating room, nor on those whom no reasonable surgeon would have incised. Instead, this analysis aimed to address scenarios of uncertainty, in which the surgeon, or a surgeon, may have had both courses of action available, and may have wondered, “should I operate, or should I watch?”

## Methods

### Data source and population:

We included patients with a principal diagnosis for an emergency general surgery condition who were ≥ 65.5 years of age, enrolled in Medicare fee-for-service and admitted through the emergency department to a nonfederal, acute care hospital between July 1, 2015 and June 30, 2018. The coding strategy is detailed in Supplemental Section 1. Emergency general surgery conditions were identified using International Classification of Diseases, Ninth or Tenth Revision, Clinical Modification codes (ICD-9/10-CM). Dates and ages allowed a 6-month “look-back” period to confirm the index admission and gather information regarding comorbidities. Patients were classified into one of 5 abdominal emergency general surgery condition categories based on principle diagnosis codes: colorectal (e.g. diverticulitis, volvulus), general abdominal (e.g. peritonitis, abdominal pain), hepatopancreaticobiliary (e.g. cholecystitis, liver abscess), intestinal obstruction (e.g. small bowel obstruction, intussusception), and upper gastrointestinal (e.g. peptic ulcer disease, appendicitis). Cardiac, thoracic, soft tissue, and vascular conditions were excluded. Transfer patients were excluded. Diagnoses included in each category are drawn from Shafi et al.[11] and are shown in Supplemental Table 1. Surgeons were included if they performed ≥ 5 general surgery operations per year for Medicare patients to allow for valid assessment of the instrumental variable. This threshold was chosen to maximize our inclusion of the population of older adults with EGS conditions, and is comparable to other large studies of surgical outcomes.[12,13] Medicare patients account for 1/3 to 1/2 of EGS patients.[7,11,14] General surgery operations were identified using standard CPT codes.[15] Patients who did not have

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continuous Part A & B coverage or who were enrolled in an HMO at any time between 6 months prior to and 6 months after their index EGS admission were excluded to ensure that available claims represented all billable encounters. Finally, patients who did not have either surgery or a surgical consultation were excluded (see Supplemental Table 2). For patients undergoing multiple operations, only the first operation was studied.

### Exposure:

The exposure of interest was operative management, defined using ICD-CM-9/10 procedure codes and Current Procedural Terminology (CPT) codes. As is customary in studies focused on emergency general surgery,[2,11,16–18] patients with qualifying operations on hospital day 0,1, or 2 were assigned to the operative treatment group.

### Outcomes:

Outcomes of interest included mortality in-hospital and at 30, 90, and 180 days. Readmissions were tracked at 30, 90, and 180 days. Additional outcomes were complications, including post-operative complications for which operative management is a prerequisite (e.g. post-operative hemorrhage). Complications were defined using ICD-CM-9/10 codes that were not present on admission (Supplemental Table 3). We also examined non-routine discharge, defined as discharge other than home without services.

### Covariates:

Claims were queried for diagnosis and procedure codes and dates, discharge destination, gender, race, zip code, and dual-eligibility status (as a proxy for socioeconomic status). Elixhauser comorbidities[19] were assessed at the index hospitalization and a 6-month look-back period. The presence of sepsis is a particularly grave comorbidity for emergency general surgery patients and was incorporated into risk adjustment models.[20] Frailty is a critical concern in the elderly and was calculated from ICD-9/10-CM codes according to the method of Kim and Schneeweiss.[21]

### Instrumental variable analysis:

In the absence of randomization, observational studies may be plagued by selection bias. In emergency general surgery conditions, patients and clinicians at the bedside have information about the patient's condition not represented in claims data. For example, patients who are more acutely ill may be more likely to undergo an emergent operation, biasing results in favor of non-operative management. In the presence of unmeasured confounding, instrumental variables can provide consistent estimates of treatment effect.[22] An instrumental variable is one that is associated with the exposure of interest (operative management) but has no direct effect on the outcome and is independent of unmeasured confounding variables. The instrumental variable serves an arbitrary nudge for some patients to receive operative care. For this analysis, we drew on methods from the pharmacoepidemiology literature[23] to derive a preference-based instrumental variable (PBIV) based on the surgeons' preferences for operative management, as we have previously described.[24] To our knowledge, this is the first US study to use a preference-based IV to assess surgical therapy.

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The instrumental variable estimate is the average causal effect for patients whose management is dertemined at least in part by the surgeon's tendency to operate (the instrumental variable). These patients are termed "compliers". For patients who would always or never receive surgery regardless of the surgeon who saw them, the instrumental variable has no effect. In the IV analysis, these pateints are given a weight of zero. Thus, the IV estimate is a weighted average that only weights results for patients for whom different surgeons would have chosen different management

To construct the PBIV, a surgeon of record (the first surgeon, in case of multiples) was identified for each operatively managed patient. For patients managed non-operatively, the attending surgeon was selected, or, if the attending of record was a non-surgeon, the first surgeon who billed for a consultation was assigned. For each surgeon, we then selected all emergency general surgery patients managed operatively or non-operatively. The final dataset was randomly split into a 10% sample to develop the PBIV and a 90% analytic set for the completion of the study to minimize bias in the instrument.[25] The instrumental variable was defined as the ratio of operative to total emergency general surgery patients for each surgeon.

A key assumption of instrumental variable analysis is ignorability. That is, patients should be assigned to surgeons with high or low preferences for operative management as-if at random. To probe this assumption, we tested balance of observable covariates across levels of the PBIV using standardized differences.[22] The IV must also be strongly associated with the exposure. We compared patients' receipt of operative management when treated by physicians above and below the median PBIV value and conducted a weak instrument test.[26] To strengthen the IV, we used near-far matching, minimizes differences in observed confounders while maximizing the contrast in IV values.[27] Patients with low values of the PBIV were matched to patients with high values on observed confounders. Patients were exact matched within hospitals, and were matched on comorbidities, condition category, patient demographics, and surgeon characteristics, as previously reported.[24] Risk adjustment was performed using methods developed by Silber.[28,29] We estimated risk differences by applying the Wald estimator in two-stage least squares after matching. Standard errors were corrected for clustering within surgeons. A robustness check was performed for outcomes that may be affected by censoring due to death. Sensitivity analyses and their results are included in the supplement in Section 3 and Table 6.

Data was analyzed using Stata Version 16.1 (StataCorp, College Station, TX) and R (R Core Team, 2017). This study was approved by the Institutional Review Board of the University of Pennsylvania (protocol number 832059).

## Results

A total of 507,677 patients met inclusion criteria, including 155,653 who had an operation (30.7%). Mean age was 74 years. Differences between patients undergoing operative management and non-operative management were generally subtle, prior to matching and clinically and statistically insignificant after matching as shown in Table 1 and Supplemental Table 4. For example, operative patients had a mean of 3.3 comorbidities compared to

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3.6 for nonoperative prior to matching and both groups had a mean of 3.4 comorbidities after matching. Characteristics of patients before and after matching in the operative and nonoperative groups by condition type are included in Supplemental Table 5.

## Unadjusted outcomes

The overall, unadjusted inpatient mortality rate was 2.2% for colorectal conditions, 5.9% for general abdominal conditions, 1.2% for hepatopancreaticobiliary conditions, 3.0% for intestinal obstruction, and 4.1% for upper GI conditions. In unadjusted analysis, operative management was associated with higher rates of inpatient mortality for colorectal conditions, general abdominal conditions, and intestinal obstruction. Mortality was lower in the operative group for hepatopancreaticobiliary conditions and there was no difference in in-hospital mortality for upper GI conditions. For colorectal, general abdominal conditions, and resuscitation, this relationship persisted through 180 days. For hepatopancreaticobiliary conditions, mortality remained lower with operative management at 180 days. For intestinal obstruction, and upper GI conditions, operative management was associated with lower mortality at 30, 90, and 180 days. Mortality results at each time point are shown in Figure 1 by condition group. Readmissions, complications, and discharge disposition for each condition group are shown in Table 2.

## Instrumental variable characteristics

The median value of the PBIV was 0.30 (IQR 0.21, 0.39). The PBIV was strongly associated with operative management. The F-statistic from a weak instrument test was 22,067, above the 10% critical value of 16. The percent of patients receiving operative management was 13% for the lowest PBIV quartile, 16% for the second quartile, 34% for the third, and 47% for the fourth. Matches were found for 224,350 of 507,677 emergency general surgery patients (44.2%). Prior to matching the largest standardized difference for any covariate was 0.19 After matching, all standardized differences were ≤0.08, well below that accepted threshold of 0.10. Additional detail on IV derivation and characteristics can be found in Supplemental Section 5.

## Adjusted outcomes

The results for mortality at each point in time are shown in Figure 2 by condition group. Results for complications, discharge disposition, and readmissions are shown in Table 3.

**Hepatopancreaticobiliary conditions:** Among 60,094 patients with hepatopancreaticobiliary conditions, operative management was not associated with any difference in in-hospital mortality, but carried a lower risk of mortality at 30 (2.6% [−4.0, −1.3]), 90 (−4.7% [−6.50, −2.8]), and 180 (−6.4% [−8.5, −4.2]) days. Operative management had lower risk of nonroutine discharge and readmissions at 30 and 90 days.

**Intestinal obstruction:** Among 56,582 patients with intestinal obstruction, operative management was associated with higher risk of mortality in the hospital (2.8% [0.7, 4.9]), no significant risk difference at 30 days, and lower, but not statistically different, mortality at 90 and 180 days. Readmissions were significantly lower after operative management at 30 and 90 days (−9.2%, [−14.2, −4.2] and −13.3% [−19.0, −7.6], respectively). Risk of

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complications was higher for operative management (18.6% [13.8, 23.4]), as was the risk of non-routine discharge (15.4% [0.1, 21.7]).

**Upper GI conditions:** Among 41,878 patients with upper gastrointestinal conditions, operative management was associated with a 9.7% higher risk of death in the hospital (6.4, 13.1), which rose to 12.7% (8.1, 13.1) at 30 days, 15.9% (10.2, 21.6) at 90 days, and 12.6% (6.4, 18.8) at 180 days. Operative management was also associated with higher risk of complications (28.7% [22.1, 35.4]), and nonroutine discharge (25.8% [17.4, 34.1]) and equivalent risk of readmission at 30 and 90 days.

**Colorectal conditions:** We included 33,326 patients with colorectal conditions. Operative management was associated with 6.9% higher risk of mortality in the hospital (95% CI 3.6, 10.2), at 30 days (5.5 [0.3, 10.6]), and at 90 days (7.4 [0.7, 14.1]), but equivalent mortality at 180 days (2.7 [−4.9, 10.3]). Operative patients also had 33.4% higher risk of complications (95% CI 24.8, 42.0) and of nonroutine discharges (53.1 [42.0, 6.3]). There was no significant difference in readmissions at 30 or 90 days.

**General abdominal conditions:** There were 32,470 patients with general abdominal conditions. Operative management was associated with 12.2% increased risk of inpatient mortality (95% CI 8.7, 15.8). This effect was attenuated at 30 days (8.5% increased risk, 95% CI 3.8, 13.2) and was nonsignificant at 90 and 180 days. Operative management was associated with 32.9% (95% CI 26.7, 39.1) higher risk of complications and 25.7% (18.0, 33.4) higher risk of nonroutine discharge. Operative management was associated with lower risk of readmission at 30 (−10.5% [−17.2, −3.8]) and 90 days (−14.9% [−22.3, −7.6]).

## Discussion

It is often said that the best surgeons are not the ones who know how to operate, but those who know when to operate. In emergency general surgery, these decisions are often fraught with uncertainty. Older adults who are often acutely and chronically ill may present with a condition that can be managed operatively or nonoperatively, with little evidence available to guide shared decision-making. These decisions are critical, as emergency general surgery conditions are responsible for 7% of all hospitalizations, 800,000 operations, and $28 billion in healthcare costs each year.[4,30,31] Moreover, variation in care for emergency general surgery conditions accounts for approximately 20,000 excess deaths each year,[3,32] with wide variation in care for older adults.[33] In this national study of older adults, we used an instrumental variable analysis to both identify the key population for whom non-clinical circumstances might dictate the course of management, and to address major challenges of unmeasured confounding. Consistent with prior evidence, we found that the impact of operative management varied widely among emergency general surgery conditions.[34]

We expected that operative management would be associated with higher risk of inpatient mortality and complications, in keeping with the findings of Shah et al.[7], but that benefit might accrue over time to yield lower risk of long-term mortality and readmissions. Hepatopancreaticobiliary conditions approximated this pattern, with operative management showing no difference for in-hospital mortality, but lower risk over time, along with

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lower risk of readmissions. These results are in line with evidence supporting early vs. delayed cholecystectomy for acute cholecystitis, even in patients with underlying systemic illness.[35–38] For intestinal obstruction, a small increase in hospital mortality with operative management was nonsignificant after discharge, and readmissions were lower, indicating potential advantages. Surgeons contemplating the management of patients for whom both operative and nonoperative options are reasonable, should strongly consider proceeding to operation. Referring providers should equally strongly consider early involvement of the surgical team.

By contrast, for upper GI, colorectal, and general abdominal conditions, mortality associated with operative management remained significantly elevated through 180-day follow-up period. The findings for colorectal conditions reflect similar dynamics to studies touting the success of endoscopic stenting for obstructing masses with improved short term outcomes of nonoperative treatments.[39,40]

The upper GI cohort includes patients with a heterogeneous set of conditions such as appendicitis and peptic ulcer disease. We found higher rates of mortality at all time points associated with operative management. Trials focused on appendicitis have not shown mortality differences based on operative vs. nonoperative management, but have shown high rates of recurrent appendicitis and subsequent appendectomy after initial nonoperative management.[8,10] Our findings also differ from previous work that reported poor outcomes of peptic ulcer disease in older adults treated with non-operative management.[41] As such, while surgeons might propose avoiding an operation during the index admission for some of these conditions such as those of the colon and rectum, for others the decision will need to be tailored to the individual condition and patient. Surgeons should be involved early in the decision-making process, however referring physicians might encourage consideration of nonoperative management during the index admission.

Randomized controlled trials and high-quality observational studies comparing operative management to nonoperative management for emergency general surgery conditions are rare, even though nonoperative management has increased in popularity and plausibility in recent years.[42,43] Our results extend these studies to broader conditions, and may provide guidance for a wider range of decision points in emergency general surgery.

Our study design had several key strengths. Medicare claims data allowed us to follow patients longitudinally to collect post-discharge outcomes and to do so at a national scale. The close scrutiny of Medicare claims for fraud ensures accuracy of the surgeon assignments. While clinical registries are commonly used to track outcomes and quality in surgery, they typically only include a convenience sample of patients who undergo an operation and report 30-day outcomes. ACS-NSQIP expanded data collection to include non-operative patients in 2015 for acute appendicitis, acute cholecystitis and small bowel obstruction in 13 hospitals, but this small pilot program cannot address the full panoply of emergency general surgery conditions.[44] Data relying on billing codes is inherently limited by the accuracy of the coding process. Our list of included codes was based on Centers for Medicare and Medicaid Services General Equivalency Mapping, supplemented by literature review, and evaluated by a panel of senior surgeons. This process yielded an inclusive list of

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codes similar to that which has recently been endorsed by the American Association for the Surgery of Trauma.[14] The goal of this study was to reflect the initial management decision to the best of our ability. Patients undergoing non-operative management as Plan A may certainly have crossed over and undergone surgery if their condition changed. This approach is analogous to an intention-to-treat analysis for a randomized trial. Patients undergoing EGS operations remote from admission follow different trajectories from those undergoing early surgery, suggesting that different dynamics are at play.[18]

### Limitations

As with any research employing administrative or claims-based data, the accuracy of our work depends on coding, which may vary from case to case and institution to institution. In particular, accuracy of coding may be different between patients undergoing operative versus non-operative management, if the operation provides a more definitive diagnosis. However, the level of accuracy is commensurate with the extent of clinical knowledge on the case and is therefore unlikely to be clinically meaningful in this setting. Patients were matched within condition buckets, but the heterogenous nature of emergency general surgery conditions prevented matching on specific diagnoses. There may be additional variation within condition groups that we could not identify. These results do not consider the role of interval operations on outcomes, only the management during the index admission. While we used a preference-based instrumental variable, this analysis could not account for patient or family preferences for choice of management. Likewise, quality of life, and additional patient-centered outcomes that may be affected by operative or nonoperative management were not included here. Additional procedures such as endoscopy or interventional radiology procedures may have occurred in either group, and the availability of such adjuncts may play into surgeons' preference for operative vs. nonoperative management. This study does not provide insight into determinants of this preference, but future research in this area would be welcome.

## Conclusion

Every surgeon who treats emergencies encounters patients for whom the best path is not clear. We found that for those patients in this marginal population, outcomes varied by condition type. For colorectal and upper GI conditions, including appendicitis, outcomes may be better with nonoperative management during the index admission. For those with hepatopancreatobiliary conditions, however, operative intervention is favored, particularly over the longer term. For intestinal obstructions and general abdominal conditions, results were equivalent overall. We hope that these findings will allow improved, patient-centered, individualized discussions between surgeons, patients, and families, as well as hospital and professional policy setting.

## Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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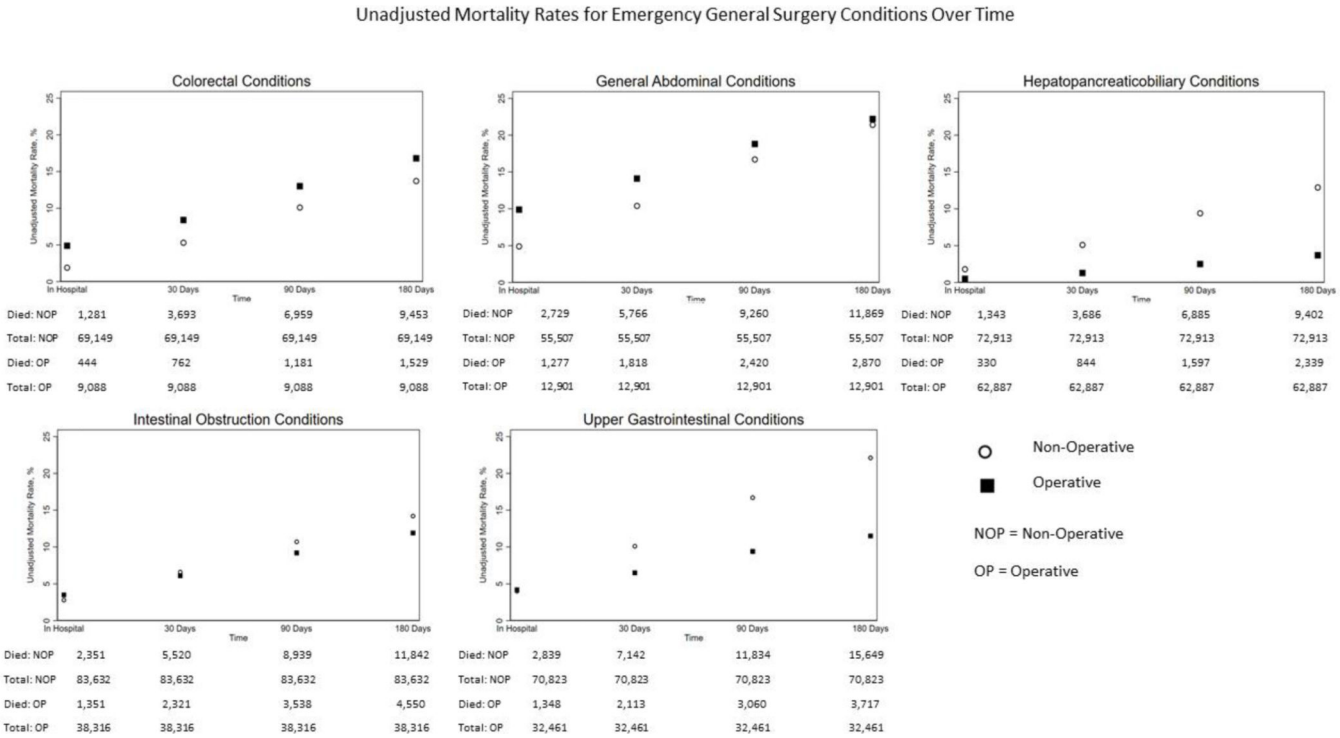


**Figure 1:**

Unadjusted mortality for operative and nonoperative emergency general surgery patients, by time interval



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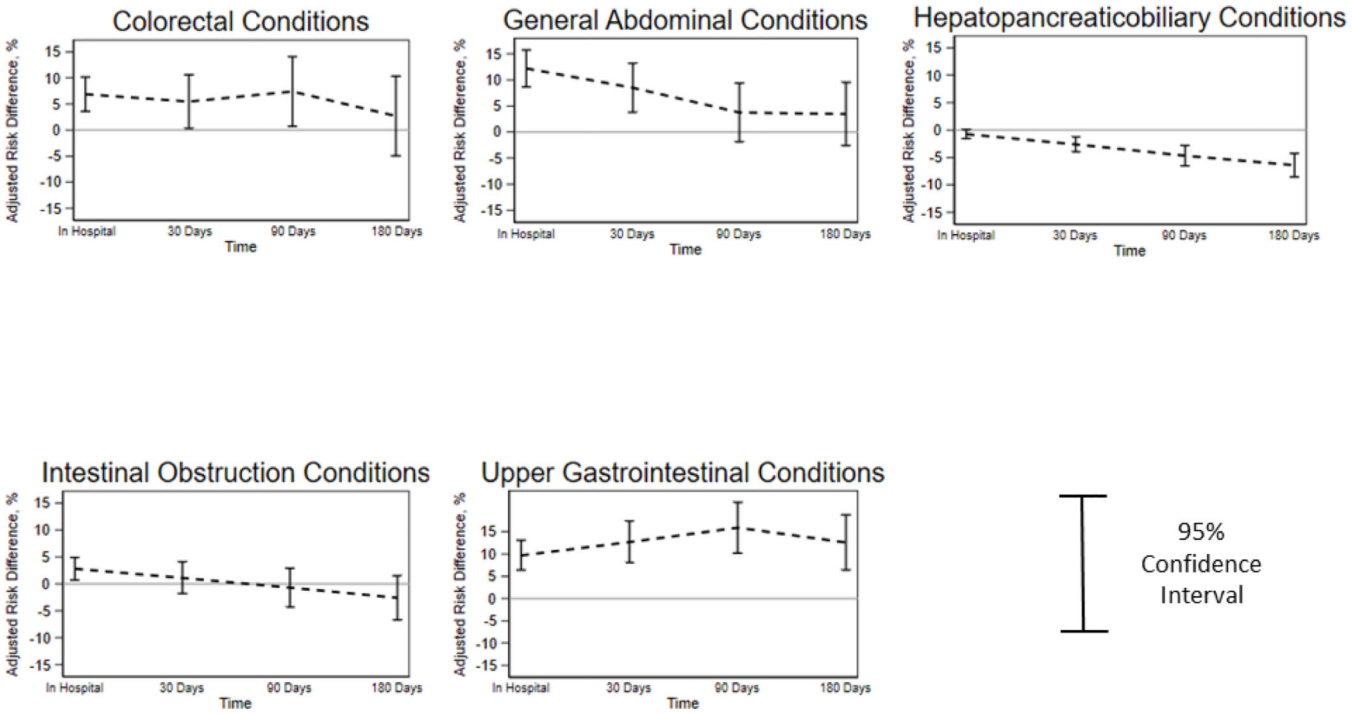


**Figure 2:**

Mortality for operative and nonoperative emergency general surgery patients, by time interval based on matched cohorts using the preference based instrumental variable.

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**Table 1.**

Selected characteristics of patients managed operatively and nonoperatively, before and after matching, selected

| | Before Matching | | | After Matching | | |
|---|---|---|---|---|---|---|
| **Covariate** | **High PBIV** | **Low PBIV** | **Std. Diff.** | **High PBIV** | **Low PBIV** | **Std. Diff.** |
| N (%) | | | | | | |
| Instrumental Variable | 0.41 (0.09) | 0.19 (0.08) | 2.51 | 0.2 (0.08) | 0.41 (0.09) | −2.44 |
| Age | 74.3 (12.03) | 74.49 (12.28) | −0.02 | 74.5 (11.95) | 74.28 (12.05) | 0.02 |
| Elixhauser Comorbidity Index | 3.34 (2.34) | 3.64 (2.49) | −0.12 | 3.34 (2.32) | 3.33 (2.33) | 0.00 |
| Predicted Risk of Death | 0.06 (0.08) | 0.07 (0.08) | −0.11 | 0.06 (0.08) | 0.06 (0.08) | 0.01 |
| Claims-Based Frailty Index | 0.16 (0.06) | 0.17 (0.06) | −0.13 | 0.16 (0.06) | 0.16 (0.06) | −0.01 |
| Preoperative Sepsis | 29668 (0.11) | 28009 (0.11) | 0 | 11688 (0.1) | 12153 (0.11) | −0.01 |
| Female Gender | 143147 (0.55) | 135640 (0.55) | −0.01 | 62230 (0.55) | 62498 (0.56) | 0.00 |
| *Race and Ethnicity* | | | | | | |
| White | 222829 (0.85) | 203320 (0.83) | 0.06 | 95610 (0.85) | 94452 (0.84) | 0.03 |
| Black | 22104 (0.08) | 28142 (0.11) | −0.1 | 10320 (0.09) | 11544 (0.1) | −0.04 |
| Other | 3519 (0.01) | 2872 (0.01) | 0.02 | 1232 (0.01) | 1312 (0.01) | −0.01 |
| Hispanic | 5722 (0.02) | 4887 (0.02) | 0.01 | 2282 (0.02) | 2248 (0.02) | 0.00 |
| Transfer Patient | 2090 (0.01) | 2469 (0.01) | −0.02 | 968 (0.01) | 1000 (0.01) | 0.00 |
| *Selected comorbidities* | | | | | | |
| Dementia | 45833 (0.17) | 49599 (0.2) | −0.07 | 19223 (0.17) | 19771 (0.18) | −0.01 |
| Chronic Kidney Disease | 111111 (0.42) | 114775 (0.47) | −0.09 | 48934 (0.44) | 47990 (0.43) | 0.01 |
| Diabetes | 110723 (0.42) | 115250 (0.47) | −0.09 | 48874 (0.44) | 49153 (0.44) | −0.01 |
| Ischemic Heart Disease | 145413 (0.55) | 148062 (0.6) | −0.1 | 64045 (0.57) | 64035 (0.57) | 0.00 |
| Cancer* | 15150 (0.06) | 14453 (0.06) | 0 | 6209 (0.06) | 6321 (0.06) | 0.00 |

*Breast, Colorectal, Lung, and Prostate

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**Table 2.**

Unadjusted Outcomes by Condition Group

| | HPB N=60,094 | Obstruction N=56,582 | Upper Gastrointestinal N=41,878 | Colorectal N=33,326 | General abdominal N=32,470 |
|---|---|---|---|---|---|
| *Complications* | | | | | |
| Operative N (%) | 8759 (13.9) | 11483 (30.0) | 8873 (27.3) | 3703 (40.7) | 5677 (44.0) |
| Non-Operative N (%) | 11105 (15.2) | 9772 (11.7) | 10851 (15.3) | 8670 (12.5) | 8674 (15.6) |
| Risk difference (95% CI) | −1.3 (−1.7, −0.9) | 18.3 (17.8,18.8) | 12.0 (11.5,12.6) | 28.2 (27.2, 29.2) | 28.4 (27.5, 29.3) |
| *Nonroutine Discharge* | | | | | |
| Operative N (%) | 14637 (23.3) | 19459 (50.8) | 12954 (39.9) | 7077 (77.9) | 8457 (65.6) |
| Non-Operative N (%) | 31963 (43.8) | 30674 (36.7) | 36239 (51.2) | 27439 (39.7) | 26833 (48.3) |
| Risk difference (95% CI) | −20.6 (−21.1, −20.1) | 14.1 (13.5,14.7) | −11.3 (−11.9, −10.6) | 38.2 (37.3,39.1) | 17.2 (16.3,18.1) |
| *Readmission 30 Days* | | | | | |
| Operative N (%) | 7478 (11.9) | 6914 (18.0) | 5616 (17.3) | 2179 (24.0) | 3253 (25.2) |
| Non-Operative N (%) | 17947 (24.6) | 16215 (19.4) | 17977 (25.4) | 15277 (22.1) | 15392 (27.7) |
| Risk difference (95% CI) | −12.7 (−13.1, −12.3) | −1.3 (−1.8, −0.9) | −8.1 (−8.6, −7.6) | 1.9 (1.0,2.8) | −2.5 (−3.4, −1.7) |

Note. HPB- Hepatopancreaticobiliary.

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**Table 3:**

Results of risk adjusted outcomes using instrumental variable, by emergency general surgery condition group

| | **HPB N=60,094** | **Obstruction N=56,582** | **Upper Gastrointestinal N=41,878** | **Colorectal N=33,326** | **General abdominal N=32,470** |
|---|---|---|---|---|---|
| Complications, risk difference (95% CI) | 2.0 (−0.9, 4.9) | 18.6 (13.8, 23.4) | 28.7 (22.1, 35.4) | 33.4 (24.8, 42.0) | 32.9 (26.7, 39.1) |
| Nonroutine Discharge, risk difference (95% CI) | −15.4 (−19.3, −11.5) | 15.4 (0.1, 21.7) | 25.8 (17.4, 34.1) | 53.1 (42.0, 6.3) | 25.7 (18.0, 33.4) |
| 30-day Readmissions, risk difference (95% CI) | −10.9 (−14.1, −7.7) | −9.2 (−14.2, −4.2) | 0.4 (−6.5, 7.2) | −0.8 (−10.6, 8.9) | −10.5 (−17.2, −3.8) |

Note. HPB- Hepatopancreaticobiliary.